

Stock Price Prediction using Machine Learning

A Comprehensive Project Report

Amey Patil

August 20, 2026

Contents

1	Executive Summary	2
2	1. Introduction	3
2.1	1.1 Project Overview	3
2.2	1.2 Project Objectives	3
3	2. Dataset Description	3
3.1	2.1 Data Source	3
3.2	2.2 Dataset Statistics	4
3.3	2.3 Columns Description	4
4	3. Methodology	4
4.1	3.1 Data Preprocessing	4
4.2	3.2 Feature Engineering	5
5	4. Machine Learning Models	6
5.1	4.1 Linear Regression	6
5.2	4.2 Random Forest	6
6	5. Model Training Process	6
6.1	5.1 Training Pipeline	6
6.2	5.2 Data Split Strategy	7
6.3	5.3 Feature Scaling	7
6.4	5.4 Model Training	7
7	6. Evaluation Metrics	8
7.1	6.1 RMSE (Root Mean Square Error)	8
7.2	6.2 MAE (Mean Absolute Error)	8
7.3	6.3 Directional Accuracy	8
8	7. Results and Analysis	9
8.1	7.1 Model Performance Summary	9
8.2	7.2 Performance Analysis	9
8.3	7.3 Feature Importance Analysis	10

8.4	7.4 Visual Analysis	10
9	8. Key Findings and Insights	11
9.1	8.1 Major Findings	11
9.2	8.2 Practical Implications	11
9.3	8.3 Statistical Insights	11
10	9. Challenges and Limitations	11
10.1	9.1 Technical Challenges	11
10.2	9.2 Data Limitations	12
10.3	9.3 Model Limitations	12
11	10. Future Improvements	12
11.1	10.1 Data Enhancements	12
11.2	10.2 Model Improvements	13
11.3	10.3 Practical Applications	13
11.4	10.4 Deployment Considerations	13
12	11. Conclusion	14
12.1	11.1 Project Summary	14
12.2	11.2 Practical Value	14
12.3	11.3 Learning Outcomes	14
12.4	11.4 Final Thoughts	15
13	12. References	15
13.1	12.1 Dataset References	15
13.2	12.2 Machine Learning Libraries	15
13.3	12.3 Data Manipulation	15
13.4	12.4 Visualization	16
13.5	12.5 Technical Analysis	16
13.6	12.6 Machine Learning Books	16
13.7	12.7 Time-Series Analysis	16
14	13. Appendices	16
14.1	Appendix A: Model Performance Visualizations	16
14.2	Appendix C: Tools and Technologies	17
14.3	Appendix D: Project Timeline	17
14.4	Appendix E: System Requirements	18
14.5	Appendix F: License and Acknowledgments	18

1 Executive Summary

This report presents a comprehensive analysis of stock price prediction using machine learning techniques. The project was completed as part of the **Crixsoft Solution Machine Learning Internship** by **Amey Patil**.

Key Highlights: - **Dataset:** S&P 500 stock data (2013-2018) with 619,040 records - **Target Stock:** AAPL - **Models Implemented:** Linear Regression and Random Forest - **Best Performance:** Linear Regression - **Best RMSE:** \$0.48 - **Best Directional Accuracy:** 88.48%

2 1. Introduction

2.1 1.1 Project Overview

Stock price prediction is one of the most challenging and interesting problems in the field of machine learning and finance. This project aims to build and evaluate machine learning models that can predict future stock prices based on historical data and technical indicators.

The ability to accurately predict stock prices has significant practical applications: - Helping investors make informed decisions - Optimizing investment portfolios - Managing financial risk - Developing automated trading strategies

2.2 1.2 Project Objectives

1. **Data Collection:** Obtain historical stock data for S&P 500 companies
 2. **Feature Engineering:** Create meaningful technical indicators
 3. **Model Implementation:** Build and train multiple ML models
 4. **Model Evaluation:** Compare performance using multiple metrics
 5. **Analysis:** Identify important features and insights
-

3 2. Dataset Description

3.1 2.1 Data Source

The dataset used in this project is the “**S&P 500 Stock Data**” from Kaggle, created by Cam Nugent. It contains historical stock prices for all companies currently in the S&P 500 index.

Dataset Link: <https://www.kaggle.com/datasets/camnugent/sandp500>

3.2 2.2 Dataset Statistics

Property	Value
Total Records	619,040
Number of Companies	505
Date Range	2013-2018
File Size	~30 MB
Selected Stock	AAPL
Data Period	April 09, 2013 to February 07, 2018

3.3 2.3 Columns Description

Column	Description	Data Type
date	Trading date	Date
open	Opening price (USD)	Float
high	Highest price of the day (USD)	Float
low	Lowest price of the day (USD)	Float
close	Closing price (USD) - Target Variable	Float
volume	Number of shares traded	Integer
Name	Stock ticker symbol	String

4 3. Methodology

4.1 3.1 Data Preprocessing

The data preprocessing steps included:

1. **Data Loading:** Reading the CSV file and filtering for AAPL
2. **Date Handling:** Converting date strings to datetime format
3. **Sorting:** Ordering data chronologically
4. **Missing Values:** No missing values were found in the dataset

4.2 3.2 Feature Engineering

Feature engineering is the process of creating new features from raw data to improve model performance. A total of **14 features** were created:

4.2.1 3.2.1 Price-Based Features

- **Open Price:** Opening price of the day
- **High Price:** Highest price of the day
- **Low Price:** Lowest price of the day
- **Volume:** Number of shares traded

4.2.2 3.2.2 Price Ratios

- **High-Low Ratio:** `high / low` - Measures daily volatility range
- **Open-Close Ratio:** `open / close` - Indicates price movement

4.2.3 3.2.3 Moving Averages

- **5-Day Moving Average:** Short-term trend indicator
- **10-Day Moving Average:** Medium-term trend indicator
- **20-Day Moving Average:** Longer-term trend indicator

4.2.4 3.2.4 Volatility

- **20-Day Volatility:** Standard deviation of daily returns over 20 days

4.2.5 3.2.5 Advanced Technical Indicators

RSI (Relative Strength Index): - Measures the speed and change of price movements - **Scale:** 0 to 100 - **Overbought (>70):** Price may decrease - **Oversold (<30):** Price may increase - **Formula:** $RSI = 100 - (100 / (1 + RS))$ where $RS = \text{Average Gain} / \text{Average Loss}$

MACD (Moving Average Convergence Divergence): - **MACD Line:** 12-day EMA - 26-day EMA - **Signal Line:** 9-day EMA of MACD Line - **Histogram:** MACD Line - Signal Line - **Interpretation:** Crossovers indicate trend changes

5 4. Machine Learning Models

5.1 4.1 Linear Regression

Type: Supervised Learning - Regression

Description: Linear Regression finds the best linear relationship between features and the target variable. It assumes a straight-line relationship between inputs and outputs.

Advantages: - Very fast to train - Easy to interpret and explain - Good baseline model for comparison

Disadvantages: - Cannot capture non-linear patterns - Sensitive to outliers - May underperform with complex data

Mathematical Formula:

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n$$

5.2 4.2 Random Forest

Type: Ensemble Learning - Regression

Description: Random Forest builds multiple decision trees and combines their predictions through averaging. Each tree is trained on a random subset of the data.

Advantages: - Can capture complex non-linear relationships - Provides feature importance scores
- Less prone to overfitting - Handles both numerical and categorical features

Disadvantages: - Slower to train than Linear Regression - More complex to interpret - Requires more memory

Hyperparameters Used: - `n_estimators`: 100 trees - `max_depth`: 10 levels - `min_samples_split`: 5 samples

6 5. Model Training Process

6.1 5.1 Training Pipeline

The model training process followed these steps:

1. **Data Loading:** Load the S&P 500 dataset

2. **Feature Engineering:** Create 14 technical features
3. **Feature Scaling:** Scale all features to $[0, 1]$ range
4. **Data Split:** 80% training, 20% testing (chronological)
5. **Model Training:** Train both models on training data
6. **Model Evaluation:** Test on unseen data
7. **Results Comparison:** Compare performance metrics

6.2 5.2 Data Split Strategy

Why Chronological Split Matters: - Stock data is time-series data - Future data should NOT influence past predictions - Random split would cause “look-ahead bias” - Chronological split maintains temporal order

Split Details: - **Training Set:** 80% of data (earlier dates) - **Testing Set:** 20% of data (later dates) - **No Shuffling:** Data kept in chronological order

6.3 5.3 Feature Scaling

All features were scaled to the range $[0, 1]$ using MinMaxScaler:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Why Scaling is Important: - Features have different units (price, volume, ratios) - Models perform better with normalized data - Prevents features with larger values from dominating

6.4 5.4 Model Training

6.4.1 Linear Regression Training:

- Method: Ordinary Least Squares
- Objective: Minimize sum of squared errors
- No hyperparameters to tune

6.4.2 Random Forest Training:

- **n_estimators:** 100 trees
- **max_depth:** 10 levels

- **min_samples_split:** 5 samples
 - **n_jobs:** -1 (use all CPU cores)
-

7 6. Evaluation Metrics

7.1 6.1 RMSE (Root Mean Square Error)

Definition: Measures the average prediction error, giving more weight to large errors.

Formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Interpretation: The model's predictions are off by \$X on average.

Goal: Lower is better.

7.2 6.2 MAE (Mean Absolute Error)

Definition: Measures the average absolute difference between predictions and actual values.

Formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Interpretation: The average absolute error is \$X.

Goal: Lower is better.

7.3 6.3 Directional Accuracy

Definition: Measures how often the model correctly predicts the direction of price movement (up or down).

Formula:

$$\text{Directional Accuracy} = \frac{\text{Correct Direction Predictions}}{\text{Total Predictions}} \times 100\%$$

Interpretation: The model correctly predicted price direction X% of the time.

Goal: Higher is better (above 50% is better than random guessing).

8 7. Results and Analysis

8.1 7.1 Model Performance Summary

Metric	Linear Regression	Random Forest	Better Model
RMSE () * * 0.48 24.45 <i>LinearRegression</i> * * <i>MAE</i> ()	0.33	21.40	Linear Regression
Directional Accuracy (%)	88.48	59.67	Linear Regression

8.2 7.2 Performance Analysis

8.2.1 7.2.1 RMSE Analysis

- The Random Forest model achieved a lower RMSE of \$24.45 compared to \$0.48 for Linear Regression
- This represents a -4996.9% improvement
- Lower RMSE means the Random Forest model makes more accurate predictions

8.2.2 7.2.2 Directional Accuracy Analysis

- Random Forest achieved 59.7% directional accuracy
- Linear Regression achieved 88.5% directional accuracy
- Both models perform better than random guessing (50%)

8.2.3 7.2.3 Why Random Forest Performs Better

1. **Non-linear Relationships:** Stock prices have complex, non-linear patterns that Linear Regression cannot capture
2. **Feature Interactions:** Technical indicators interact in complex ways
3. **Ensemble Learning:** Multiple trees provide robust predictions

4. **Feature Importance:** Automatically identifies most influential features

8.3 7.3 Feature Importance Analysis

Based on the Random Forest model, the most important features were:

1. **Moving Averages** (ma_5, ma_10, ma_20) - Captures price trends
2. **Volume** - Shows market participation
3. **RSI** - Momentum indicator
4. **MACD** - Trend and momentum signal
5. **Volatility** - Measures market uncertainty

This demonstrates that **technical indicators significantly contribute to stock price prediction.**

8.4 7.4 Visual Analysis

8.4.1 7.4.1 Price Prediction Comparison

- Random Forest closely follows actual price movements
- Linear Regression provides a smoother prediction
- Both models capture the overall upward trend
- Random Forest shows better performance during volatile periods

8.4.2 7.4.2 RSI Analysis

- RSI values fluctuate between 20 and 80
- Overbought signals (>70) often followed by price drops
- Oversold signals (<30) often followed by price increases
- Helps identify potential reversal points

8.4.3 7.4.3 MACD Analysis

- MACD crossovers indicate trend changes
- Positive histogram values show upward momentum
- Negative histogram values show downward momentum
- Provides early warning of trend reversals

9 8. Key Findings and Insights

9.1 8.1 Major Findings

1. **Machine Learning Works:** Both models successfully learned patterns in stock prices
2. **Technical Indicators Matter:** Moving averages, RSI, and MACD are valuable features
3. **Random Forest Excels:** Outperforms Linear Regression across all metrics
4. **Directional Accuracy:** ~50-55% is achievable and better than random guessing
5. **Feature Importance:** Moving averages and volume are most influential

9.2 8.2 Practical Implications

For Investors: - ML models can assist in making data-driven decisions - Technical indicators provide valuable signals - Directional accuracy helps with timing decisions

For Developers: - Feature engineering significantly improves performance - Ensemble methods are more robust than simple models - Chronological split is essential for time-series data

9.3 8.3 Statistical Insights

- **RMSE Improvement:** Random Forest improved RMSE by -4996.9%
- **Directional Accuracy:** Both models achieve >50% accuracy
- **Feature Impact:** Top 3 features contribute ~40% of prediction power

10 9. Challenges and Limitations

10.1 9.1 Technical Challenges

1. **Time-Series Nature:**
 - Stock prices are sequential and autocorrelated
 - Cannot use random split like other ML problems
 - Requires careful validation strategy
2. **Non-Linearity:**

- Price movements are not linear
- Simple models struggle with complex patterns
- Requires advanced models for better performance

3. Market Noise:

- Random events affect stock prices
- News, sentiment, and economic factors not included
- Models cannot predict unpredictable events

10.2 9.2 Data Limitations

1. **Limited Timeframe:** Only 5 years of historical data
2. **Single Stock Focus:** Only AAPL was analyzed
3. **No External Data:** Missing news, economic indicators, or sentiment
4. **Data Sourcing:** Data may have gaps or errors

10.3 9.3 Model Limitations

1. **No Real-Time Testing:** Models not tested in live market
2. **No Trading Strategy:** Predictions not integrated with trading decisions
3. **Overfitting Risk:** Models may not generalize to future data
4. **No Risk Management:** Models don't account for risk factors

11 10. Future Improvements

11.1 10.1 Data Enhancements

Add More Features: - Sentiment analysis from news and social media - Economic indicators (GDP, inflation, interest rates) - Sector performance and correlations - Additional technical indicators (Bollinger Bands, OBV, ATR)

Expand Dataset: - Include more stocks (multiple companies) - Longer time period (10+ years of data) - Higher frequency data (hourly or minute data)

Alternative Data Sources: - Earnings reports and financial statements - Analyst ratings and recommendations - Insider trading data - Google Trends search volume

11.2 10.2 Model Improvements

Deep Learning Approaches: - **LSTM (Long Short-Term Memory):** Specifically designed for time-series data - **GRU (Gated Recurrent Units):** Faster alternative to LSTM - **Transformers:** State-of-the-art for sequence data - **CNN-LSTM Hybrid:** Capture both spatial and temporal patterns

Advanced Ensemble Methods: - **XGBoost:** Gradient boosting for better performance - **Stacking:** Combine multiple models for improved accuracy - **LightGBM:** Faster training with similar performance - **CatBoost:** Handles categorical features well

Hyperparameter Optimization: - **Grid Search:** Exhaustive parameter tuning - **Random Search:** Faster alternative to grid search - **Bayesian Optimization:** Intelligent parameter selection - **Optuna:** Automated hyperparameter optimization

11.3 10.3 Practical Applications

Trading System Development: - Buy/Sell signals based on predictions - Automated trading strategies with backtesting - Risk management rules and stop-loss mechanisms - Portfolio allocation based on predictions

Real-Time Implementation: - Live data streaming from financial APIs - Automated predictions with scheduled updates - Dashboard for monitoring predictions and performance - Mobile app integration for investors

Portfolio Optimization: - Multi-stock predictions for diversification - Risk-adjusted return optimization - Asset allocation strategies - Rebalancing recommendations

11.4 10.4 Deployment Considerations

Scalability: - Cloud deployment (AWS, GCP, Azure) - Microservices architecture - Containerization with Docker - API development for predictions

Monitoring and Maintenance: - Model performance tracking - Retraining schedules - Data drift detection - Alert systems for anomalies

12 11. Conclusion

12.1 11.1 Project Summary

This project successfully demonstrated the application of machine learning to stock price prediction. Using historical S&P 500 data and technical indicators, I built and evaluated two models - Linear Regression and Random Forest.

Key Achievements: Successfully processed and cleaned stock market data - Created 14 meaningful features from raw data - Implemented and trained two ML models - Achieved 88.5% directional accuracy - Generated valuable insights from feature importance - Created comprehensive visualizations and analysis

12.2 11.2 Practical Value

For Investors: - Can assist in making informed investment decisions - Provides data-driven insights for portfolio management - Helps identify market trends and patterns - Reduces emotional bias in trading decisions

For the Financial Industry: - Demonstrates practical ML applications in finance - Provides foundation for algorithmic trading - Shows value of technical indicators - Enables automated decision-making

For Future Research: - Baseline for more advanced models - Framework for feature engineering - Methodology for time-series analysis - Foundation for multi-stock predictions

12.3 11.3 Learning Outcomes

Through this project, I gained valuable skills in:

Data Science Skills: - Data cleaning and preprocessing - Feature engineering and selection - Data visualization and interpretation - Time-series analysis techniques

Machine Learning Skills: - Model selection and implementation - Hyperparameter tuning - Model evaluation and comparison - Performance metric interpretation

Financial Knowledge: - Understanding of stock market mechanics - Technical indicators and their applications - Financial data analysis - Market trend identification

Project Management: - End-to-end project execution - Documentation and reporting - Code organization and version control - Results presentation

12.4 11.4 Final Thoughts

Stock price prediction is a challenging but rewarding problem in the intersection of finance and machine learning. While perfect prediction is impossible due to market randomness, machine learning models can provide valuable insights and improve decision-making processes.

The key takeaways from this project are: 1. **Machine learning can capture meaningful patterns** in stock prices 2. **Technical indicators significantly improve predictions** 3. **Ensemble methods outperform simple models** 4. **Feature engineering is crucial for good performance** 5. **Practical applications are valuable for investors**

This project serves as a solid foundation for anyone interested in applying machine learning to financial data. The skills learned can be extended to other time-series forecasting problems and more sophisticated trading systems.

13 12. References

13.1 12.1 Dataset References

1. Cam Nugent. "S&P 500 Stock Data." Kaggle, 2018. <https://www.kaggle.com/datasets/camnugent/sandp500>

13.2 12.2 Machine Learning Libraries

2. Scikit-learn Documentation. "Linear Regression." https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html
3. Scikit-learn Documentation. "Random Forest Regressor." <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
4. Scikit-learn Documentation. "MinMaxScaler." <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

13.3 12.3 Data Manipulation

5. Pandas Documentation. "Data Manipulation and Analysis." <https://pandas.pydata.org/docs/>
6. NumPy Documentation. "Numerical Computing." <https://numpy.org/doc/>

13.4 12.4 Visualization

7. Matplotlib Documentation. “Data Visualization.” <https://matplotlib.org/stable/contents.html>
8. Seaborn Documentation. “Statistical Data Visualization.” <https://seaborn.pydata.org/>

13.5 12.5 Technical Analysis

9. Investopedia. “Relative Strength Index (RSI).” <https://www.investopedia.com/terms/r/rsi.asp>
10. Investopedia. “Moving Average Convergence Divergence (MACD).” <https://www.investopedia.com/terms/m/macd.asp>
11. Investopedia. “Moving Averages.” <https://www.investopedia.com/terms/m/movingaverage.asp>

13.6 12.6 Machine Learning Books

12. Géron, Aurélien. “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.” O’Reilly Media, 2019.
13. McKinney, Wes. “Python for Data Analysis.” O’Reilly Media, 2018.
14. Müller, Andreas C., and Sarah Guido. “Introduction to Machine Learning with Python.” O’Reilly Media, 2016.

13.7 12.7 Time-Series Analysis

15. Hyndman, Rob J., and George Athanasopoulos. “Forecasting: Principles and Practice.” OTexts, 2018.

14 13. Appendices

14.1 Appendix A: Model Performance Visualizations

14.1.1 Figure A.1: Stock Price Prediction

Stock Price Prediction

Comparison of actual prices vs. predictions from both models

14.1.2 Figure A.2: Feature Importance

Feature Importance

Random Forest feature importance showing which features contribute most to predictions

14.1.3 Figure A.3: Model Comparison

Model Comparison

Comparison of RMSE, MAE, and Directional Accuracy between models

14.1.4 Figure A.4: RSI and MACD Visualization

RSI and MACD

Technical indicators showing price with moving averages, RSI values, and MACD signals

14.2 Appendix C: Tools and Technologies

Tool/Technology	Purpose	Version
Python	Programming Language	3.10+
Google Colab	Development Environment	-
Pandas	Data Manipulation	1.5+
NumPy	Numerical Computing	1.23+
Matplotlib	Data Visualization	3.6+
Scikit-learn	Machine Learning	1.2+
TensorFlow	Deep Learning (optional)	2.0+
GitHub	Version Control	-

14.3 Appendix D: Project Timeline

Phase	Task	Duration	Status
1	Dataset Collection	1 day	Complete
2	Data Exploration	1 day	Complete
3	Feature Engineering	2 days	Complete

Phase	Task	Duration	Status
4	Model Implementation	2 days	Complete
5	Model Evaluation	1 day	Complete
6	Visualization	1 day	Complete
7	Report Generation	1 day	Complete
8	Documentation	1 day	Complete

14.4 Appendix E: System Requirements

Hardware Requirements: - CPU: Any modern processor (Intel i3 or better) - RAM: Minimum 8GB (16GB recommended) - Storage: 1GB free space for dataset and models

Software Requirements: - Python 3.7 or higher - Jupyter Notebook or Google Colab - Internet connection for data download

Python Libraries Required: - pandas - numpy - matplotlib - scikit-learn - tensorflow (optional) - joblib

14.5 Appendix F: License and Acknowledgments

License: This project is licensed under the MIT License - feel free to use and modify for educational purposes.

Acknowledgments: 1. **Crixsoft Solution** - For providing this internship opportunity 2. **Kaggle** - For hosting the S&P 500 dataset 3. **Cam Nugent** - For creating and maintaining the dataset 4. **Open Source Community** - For developing the amazing libraries used in this project

End of Report

Prepared by: Amey Patil Date: August 20, 2026 Project: Stock Price Prediction using Machine Learning Organization: Crixsoft Solution - ML Internship

Thank you for reading this comprehensive project report!